自进化

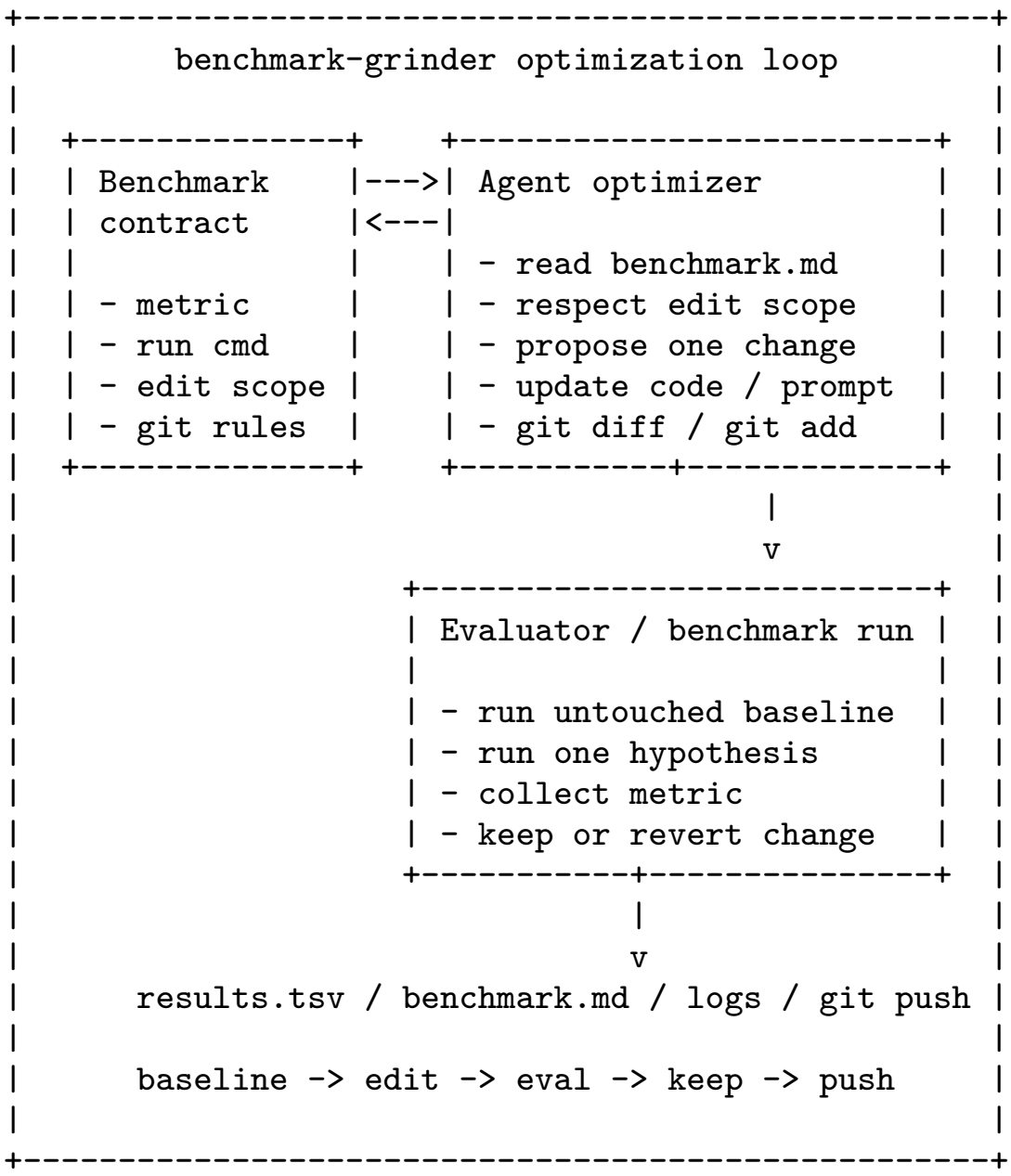
NVIDIA: AVO: Agentic Variation Operators for Autonomous Evolutionary Search

类似 alphaevolve

Karpathy: autoresearch
https://github.com/karpathy/autoresearch

为了验证其有效性，英伟达将 AVO 应用于 **NVIDIA Blackwell B200 GPU** 上的多头注意力（MHA）内核，并直接与专家优化的 cuDNN 和 FlashAttention-4 内核进行对比。在无需人工干预、长达 7 天的连续自主演化中，智能体探索了超过 500 个优化方向，演化出 40 个内核版本。最终生成的 MHA 内核在 BF16 精度下达到了最高 **1668 TFLOPS** 的吞吐量，在测试配置中分别超越 cuDNN 高达 **3.5%**，超越 FlashAttention-4 高达 **10.5%**。

benchmark-grinder skills

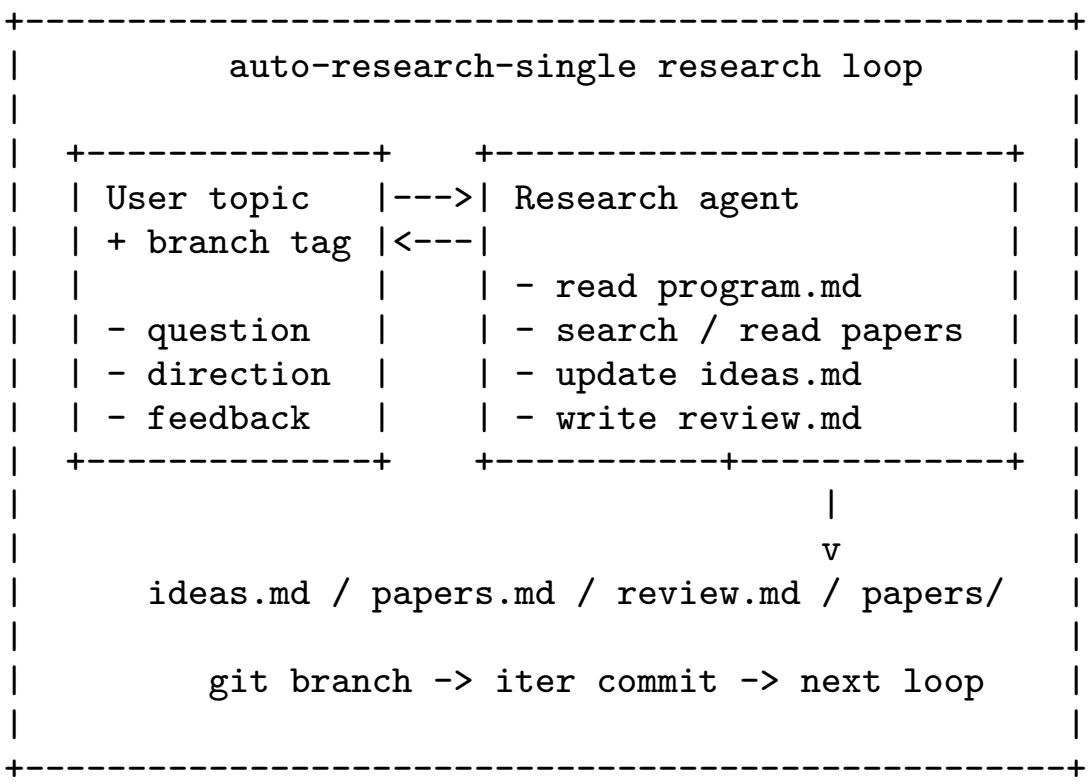


benchmark-grinder: Experiment Log

Exp	F1	Status	Key change
Baseline	29.76%	-	Original prompts
Exp1	30.07%	Kept	Category-specific prompts
Exp2	29.87%	Reverted	Enhanced temporal prompt
Exp3	N/A	Crashed	`top_k=16` (SSL error)
Exp4	29.68%	Discarded	Constraint-aware query
Exp5	22.89%	Discarded	Temporal weight boost
Exp6	29.02%	Discarded	Chain-of-Thought for Cat3

This is the intended grinder loop in practice: keep justified gains, revert regressions, and record failures.

auto-research-single: Architecture



Literature-first autonomous research, with Markdown files as the working memory.